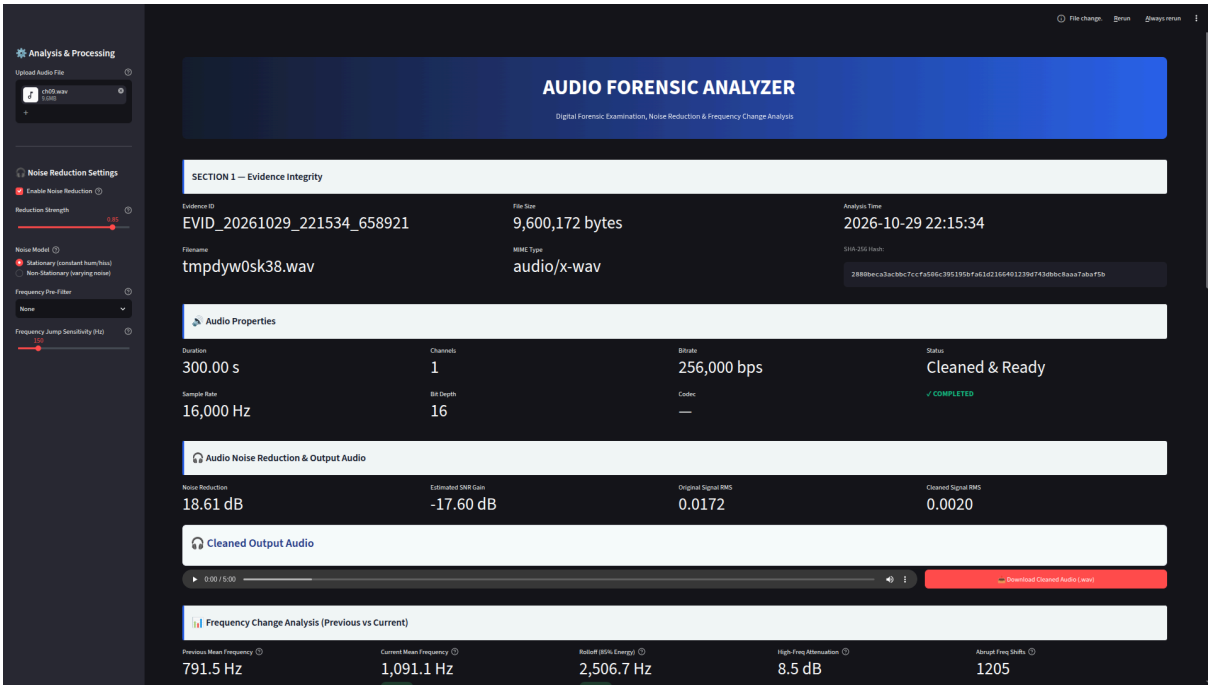
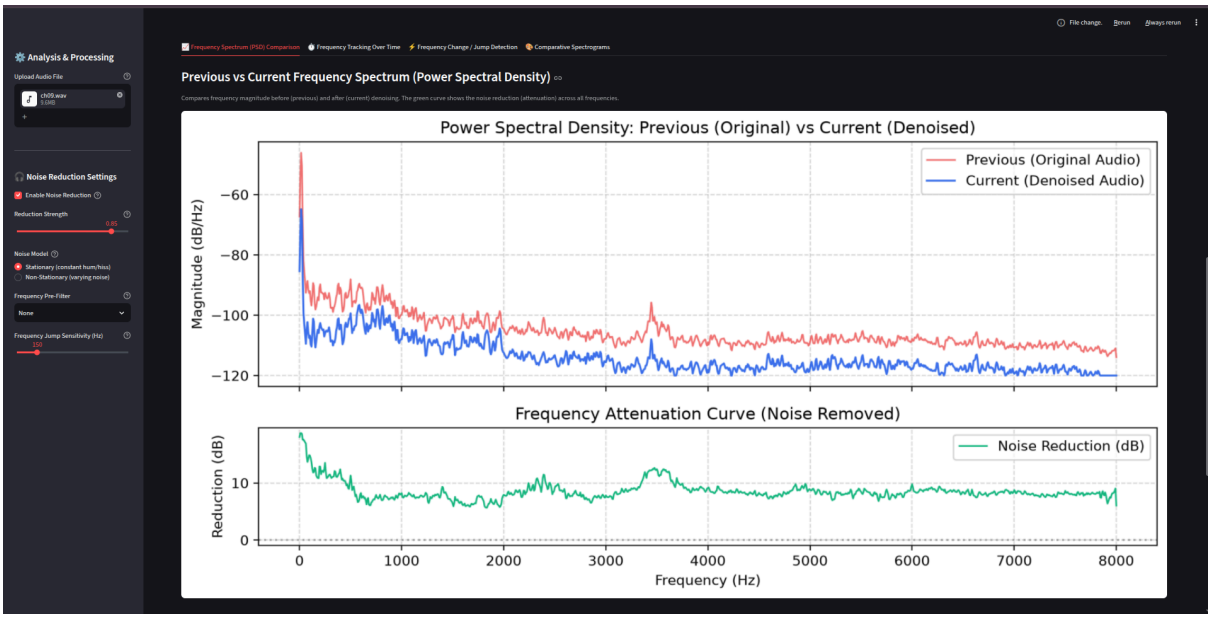
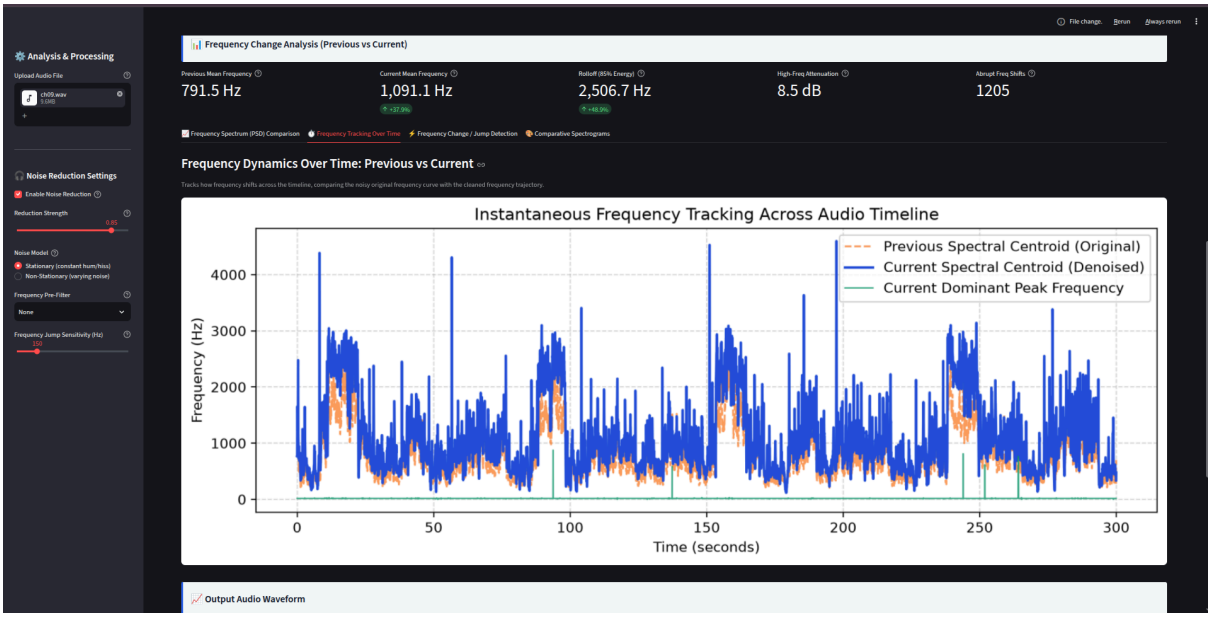
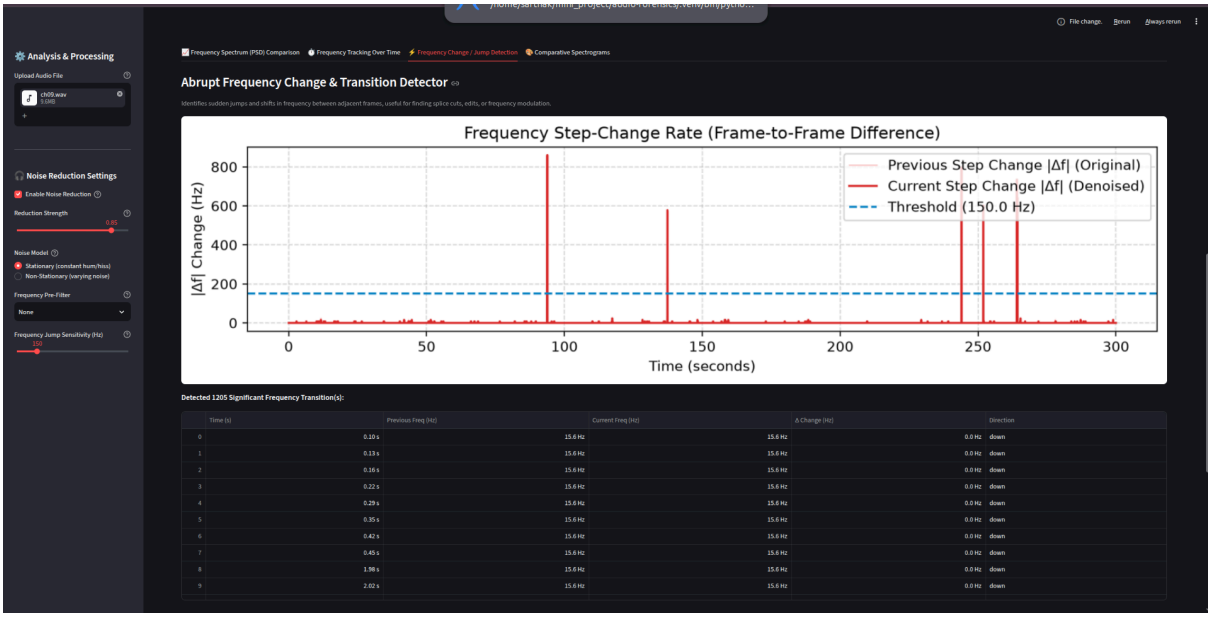
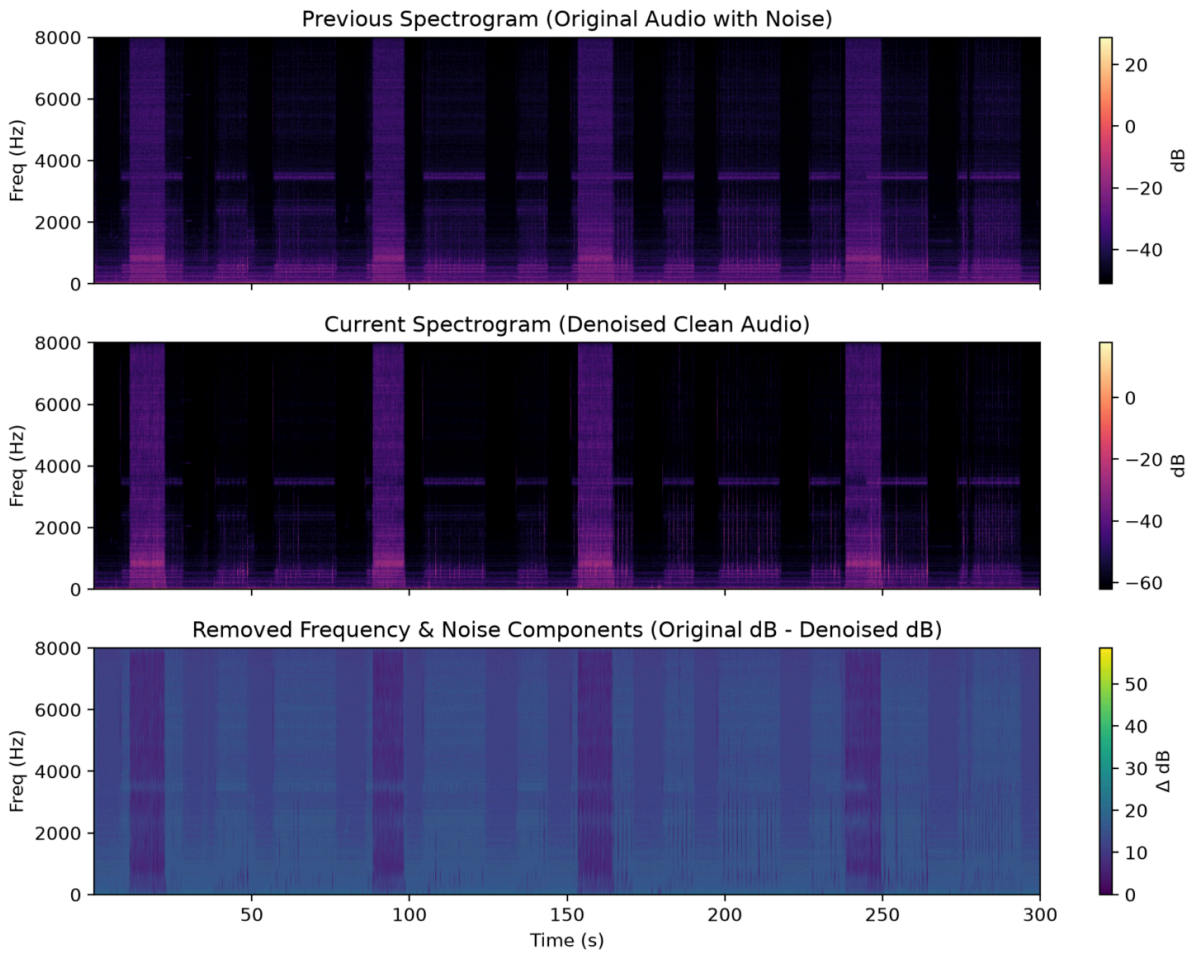

Analysis & Processing
Upload Audio File
ch09.wav
Noise Reduction Settings
Enable Noise Reduction
Reduction Strength
0.85
Noise Model
Stationary (constant hum/hiss)
Non-Stationary (varying noise)
Frequency Pre-Filter
None
Frequency Jump Sensitivity (Hz)
150
File change
Rerun
Always rerun
AUDIO FORENSIC ANALYZER
Digital Forensic Examination, Noise Reduction & Frequency Change Analysis
SECTION 1 — Evidence Integrity
Evidence ID
EVID_20261029_221534_658921
File Size
9,600,172 bytes
Analysis Time
2026-10-29 22:15:34
Filename
tmpdyw0sk38.wav
MIME Type
audio/x-wav
SHA-256 Hash:
2880beca3acbbc7ccfa506c395195bfa61d216640123 9d743dbbc8aaa7abaf5b
Audio Properties
Duration
300.00 s
Channels
1
Bitrate
256,000 bps
Status
Cleaned & Ready
Sample Rate
16,000 Hz
Bit Depth
16
Codec
—
✓ COMPLETED
Audio Noise Reduction & Output Audio
Noise Reduction
18.61 dB
Estimated SNR Gain
-17.60 dB
Original Signal RMS
0.0172
Cleaned Signal RMS
0.0020
Cleaned Output Audio
0:00 / 5:00
Download Cleaned Audio (.wav)
Frequency Change Analysis (Previous vs Current)
Previous Mean Frequency
791.5 Hz
Current Mean Frequency
1,091.1 Hz
Rolloff (85% Energy)
2,506.7 Hz
High-Freq Attenuation
8.5 dB
Abrupt Freq Shifts
1205


Analysis & Processing
Noise Reduction Settings
Frequency Spectrum (PSD) Comparison
Frequency Tracking Over Time
Frequency Change / Jump Detection
Comparative Spectrograms
Previous vs Current Frequency Spectrum (Power Spectral Density)
Compares frequency magnitude before (previous) and after (current) denoising. The green curve shows the noise reduction (attenuation) across all frequencies.
Power Spectral Density: Previous (Original) vs Current (Denoised)
Previous (Original Audio)
Current (Denoised Audio)
Magnitude (dB/Hz)
−60
−80
−100
−120
Frequency Attenuation Curve (Noise Removed)
Noise Reduction (dB)
Reduction (dB)
10
0
0
1000
2000
3000
4000
5000
6000
7000
8000
Frequency (Hz)





	Time (s)	Previous Freq (Hz)	Current Freq (Hz)	Δ Change (Hz)	Direction
0	0.10 s	15.6 Hz	15.6 Hz	0.0 Hz	down
1	0.13 s	15.6 Hz	15.6 Hz	0.0 Hz	down
2	0.16 s	15.6 Hz	15.6 Hz	0.0 Hz	down
3	0.22 s	15.6 Hz	15.6 Hz	0.0 Hz	down
4	0.29 s	15.6 Hz	15.6 Hz	0.0 Hz	down
5	0.35 s	15.6 Hz	15.6 Hz	0.0 Hz	down
6	0.42 s	15.6 Hz	15.6 Hz	0.0 Hz	down
7	0.45 s	15.6 Hz	15.6 Hz	0.0 Hz	down
8	1.98 s	15.6 Hz	15.6 Hz	0.0 Hz	down
9	2.02 s	15.6 Hz	15.6 Hz	0.0 Hz	down


Previous Spectrogram (Original Audio with Noise)
Freq (Hz)
8000
6000
4000
2000
0
20
0
−20
−40
dB
Current Spectrogram (Denoised Clean Audio)
Freq (Hz)
8000
6000
4000
2000
0
0
−20
−40
−60
dB
Removed Frequency & Noise Components (Original dB - Denoised dB)
Freq (Hz)
8000
6000
4000
2000
0
50
40
30
20
10
0
Δ dB
50
100
150
200
250
300
Time (s)